VAHID REZA KHAZAIE

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Summary

Applied scientist with 4+ years of post-MSc industry experience building large language model, NLP, and multimodal systems for business applications, and 5+ years total in machine learning research and engineering. Specialized in **core LLM technologies**: **evaluation** of foundation models at scale (benchmark design, LLM-as-a-judge, human-aligned metrics), **prompt engineering and optimization**, **fine-tuning and post-training**, and **self-supervised and self-learning** methods, with additional experience in **recommender systems** and retrieval-augmented generation. Author of **15+ peer-reviewed publications** at top-tier venues including ACM TIST, AAAI, WACV, ICASSP, MICCAI, and NeurIPS, ICML, and ICLR workshops. Experienced in owning ambiguous problems end to end: proposing evaluation frameworks and success criteria, shipping production-quality code, iterating quickly on feedback, and communicating results to technical and non-technical audiences.

Technical Skills

- **LLMs and NLP:** Transformer architectures, prompt engineering and optimization, supervised fine-tuning (SFT, LoRA/PEFT), LLM evaluation and benchmark design, LLM-as-a-judge, retrieval-augmented generation (RAG), information retrieval, agentic workflows and tool/API use, text classification, named entity recognition (NER)
- **Machine Learning:** Deep learning, self-supervised and self-learning methods, recommender systems (sequential and session-based), reinforcement learning, diffusion models, GANs, anomaly detection, computer vision, multimodal representation learning
- **Programming:** Python, C++, Bash/Shell, SQL
- **Frameworks:** PyTorch, PyTorch Lightning, Hugging Face Transformers, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib
- **Infrastructure:** Linux/Unix, Git, Docker, Slurm, multi-GPU distributed training, CI/CD, model deployment and serving
- **Analysis:** Experiment design and A/B testing, statistical hypothesis testing, large-scale data mining

Experience

Machine Learning Engineer — June 2022 – Present
Vector Institute for Artificial Intelligence — *Toronto, Canada*

- **First author** of **FairLens**, an **LLM evaluation framework** measuring vision-language model behavior on high-stakes questions across hiring, legal, and healthcare. Designed the prompt suite (69 closed- and open-ended prompts per image), proposed three complementary evaluation metrics – soundness against logical ground truth, demographic parity, and unsupported-association rate – and built an automated harness combining rule-based scoring with **LLM-as-a-judge** grading to evaluate **8 models** (GPT-5.2-reasoning, Qwen2.5-VL, Qwen3-VL, InternVL3, Ovis2.5, LLaVA-1.6, Llama-3.2-Vision, CogVLM) on **103K+ image–question pairs per model**.
- Developed **HumaniBench**, a benchmark suite evaluating **15+ multimodal LLMs** on **32K+ human-aligned image–question pairs** across seven principles including fairness, ethics, reasoning, robustness, and language inclusivity – defining evaluation criteria where no ground truth existed; **accepted to ACM Transactions on Intelligent Systems and Technology (TIST), 2026**.
- Architected a **multimodal multi-agent AI system** in which specialized LLM agents invoke **external tools and APIs** for preprocessing, analysis, verification, and mitigation across text, image, audio, video, and structured data. Designed the agent handoff and verification protocol so outputs remain auditable, supporting hiring, healthcare, and social media applications.
- Built **sequential and session-based recommender systems** (GRU4Rec, NARM) for e-commerce next-item prediction, delivering a documented, reusable open-source code base with reproducible training and evaluation pipelines for industry partners.
- Engineered **GenerativeSSL**, a **self-learning** framework combining diffusion-based generative augmentation with self-supervised learning so models improve from their own generated views without additional labels, raising downstream task accuracy by up to **10%**; published at **AAAI 2025**.
- Created **SegMate**, applying **prompt tuning and optimization** to the Segment Anything Model (PT-SAM): learned soft prompts replace manual prompting, yielding **25%+** gains in IoU, accuracy, and F1 over baseline SAM; selected as a **spotlight** at the ICLR 2024 Climate Change Workshop.
- Built and maintain **mmlearn**, an open-source toolkit for multimodal representation learning supporting **distributed multi-GPU training** and reproducible large-scale model evaluation on Linux/Slurm clusters.
- Led the **Multimodal AI** and **Anomaly Detection** programs end to end across tabular, graph, image, and video data. Ran requirements-gathering with industry stakeholders, defined success metrics, and served as technical lead for a shared code base, delivering **50 proofs of concept** with **85–90% NPS** while iterating rapidly on partner feedback.

Machine Learning Intern August 2021 – January 2022
Mitacs *London, Canada*

- Improved barcode recognition accuracy on driver's licenses by **20%** with a customized U-Net trained on a synthetic dataset generated through histogram matching and a targeted augmentation pipeline, removing the need for costly manual data collection.

Graduate Research Assistant January 2021 – June 2022
Western University *London, Canada*

- Proposed an adversarial-masked context autoencoder for unsupervised anomaly detection in which a masking network is trained adversarially against the reconstruction objective, improving AUC by **10%** over prior state of the art; published at **WACV 2022**.
- Designed an identity-preserving super-resolution network that improved face recognition on extremely low-resolution inputs (11x8 pixels) by **14%** on cumulative match characteristic over an SRGAN baseline.

Selected Publications

- **Khazaie V.R.**, Radwan A.Y., Raza S., 2026. FairLens: Benchmarking Fairness in Vision-Language Models for High-Stakes Decision-Making. (Project page)
- Raza S., Narayanan A., **Khazaie V.R.**, et al., 2026. HumaniBench: A Human-Centric Framework for Large Multimodal Models Evaluation. ACM Transactions on Intelligent Systems and Technology (TIST). (Link)
- Narayanan A., **Khazaie V.R.**, et al., 2025. Bias in the Picture: Benchmarking VLMs with Social-Cue News Images and LLM-as-Judge Assessment. NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle. (Link)
- Raval A., Narayanan A., **Khazaie V.R.**, et al., 2025. LinguaMark: Do Multimodal Models Speak Fairly? A Benchmark-Based Evaluation. NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle. (Link)
- Ayromlou S., **Khazaie V.R.**, et al., 2024. Can Generative Models Improve Self-Supervised Representation Learning? Proceedings of the AAAI Conference on Artificial Intelligence. (Link)
- Jewell J.T., **Khazaie V.R.**, et al., 2022. One-Class Learned Encoder-Decoder Network with Adversarial Context Masking for Novelty Detection. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). (Link)
- Bayat N., **Khazaie V.R.**, et al., 2021. Fast Inverse Mapping of Face GANs. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). (Link)
- Baghbanzadeh N., ..., **Khazaie V.R.**, et al., 2025. Advancing Medical Representation Learning Through High-Quality Data. International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). (Link)
- Roy S., Parhizkar Y., Ogidi F., **Khazaie V.R.**, et al., 2025. Benchmarking Vision-Language Contrastive Methods for Medical Representation Learning. ICML 2025 Workshop on Multi-modal Foundation Models and LLMs for Life Sciences. (Link)

Education

MSc in Computer Science, Western University, London, Canada January 2021 – June 2022
Thesis: Solving Challenges in Deep Unsupervised Methods for Anomaly Detection; GPA: 4.0 / 4.0
Coursework: Advanced Artificial Intelligence, Reinforcement Learning, Unstructured Data, Brain-Inspired AI

BSc in Computer Engineering, Rajaee University, Tehran, Iran September 2015 – September 2019
Coursework: Data Structures, Algorithm Design, Object-Oriented Programming, Advanced Deep Learning

Awards

Graduate Financial Package, Western University: 45.2K CAD January 2021 – April 2022
Mitacs Accelerate Fellowship: 15K CAD August 2021 – January 2022